

Hangover Cure –Searching for ALDH2 E487K Agonists

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Abstract

ALDH2 is key to alcohol metabolism, but the E487K East-Asian mutant produces an inactive enzyme. Over 100 million people are affected, but there is no effective cure. Our research analyzed Alda-1 to identify key residues for ligand binding, then screened ZINC compounds by similarity and docked them using Autodock Vina. We found 7 potential ALDH2 agonists based on whitelist and blacklist residue interaction. In the future, we plan to develop an algorithm for post-docking analysis and test our results in a wet lab setting, aiming to find foods or drugs that can be practically applied to humans, thereby curing symptoms caused by alcohol metabolism intolerance.

Motive

Many East-Asian drinkers experience alcohol flush after consumption, unlike Westerners, who can handle alcohol without visible effects. This led us to look into the E487K mutant of ALDH2, which causes enzyme inactivity. Currently, there is no effective treatment. After reviewing literature, the best approach seems to be finding agonists for ALDH2 to stabilize enzyme structure and enhance alcohol metabolism.

Goals & Methods

1. Research Goals

- Identify E487K-caused symptoms and analyze the binding conformation of existing ALDH2 agonists and inhibitors.
- Obtain docking affinity and positions using similarity ranking and Autodock Vina.
- Explore post-docking analysis based on individual ligand-residue interactions and identify potential ALDH2 agonists.
- Explore computational methods (Pair Score) to differentiate between inhibitors and agonists.

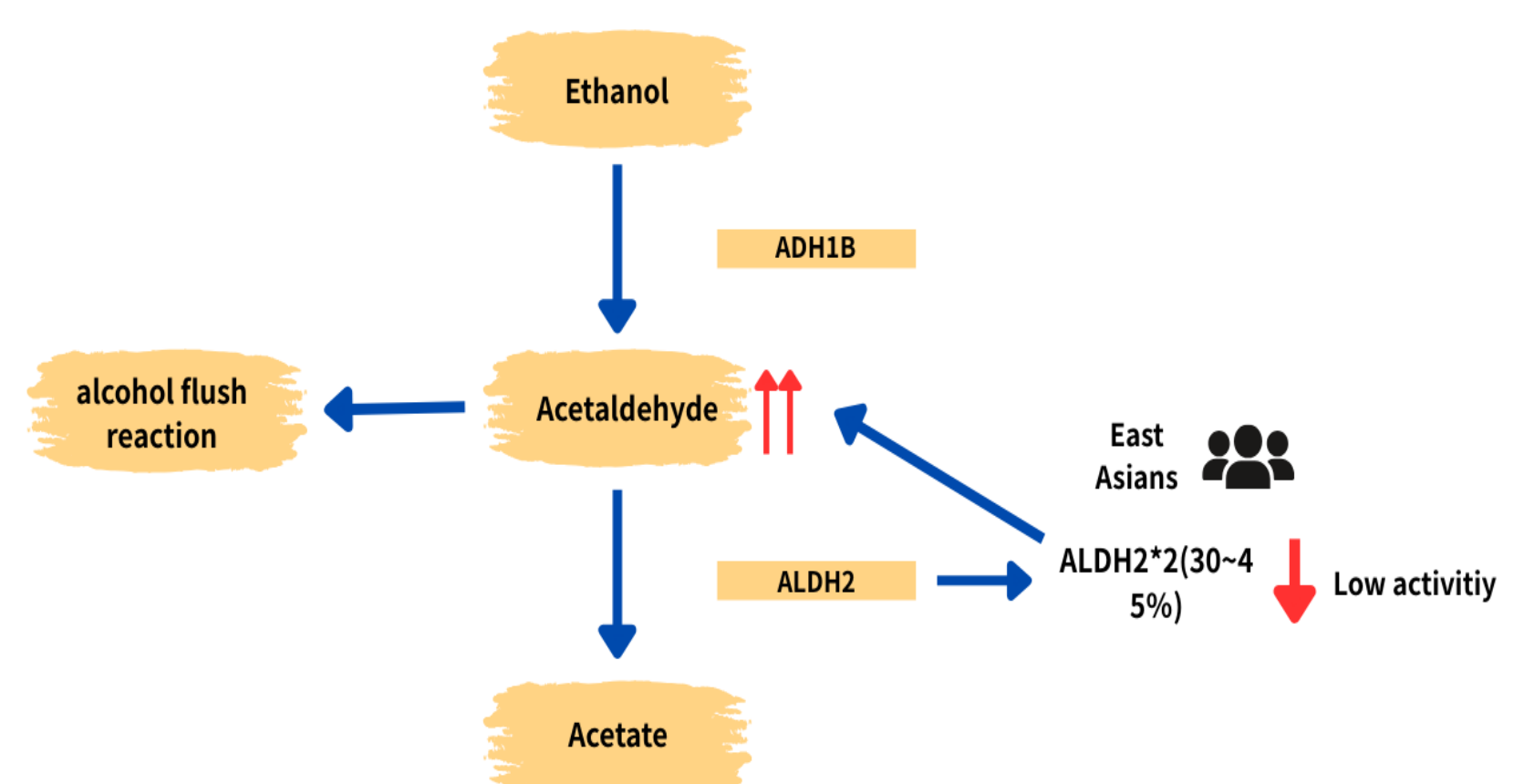


Figure 1: ALDH2 Function and Effects of E487K (ALDH2*2) Mutation

2. Computational Analysis Pipeline

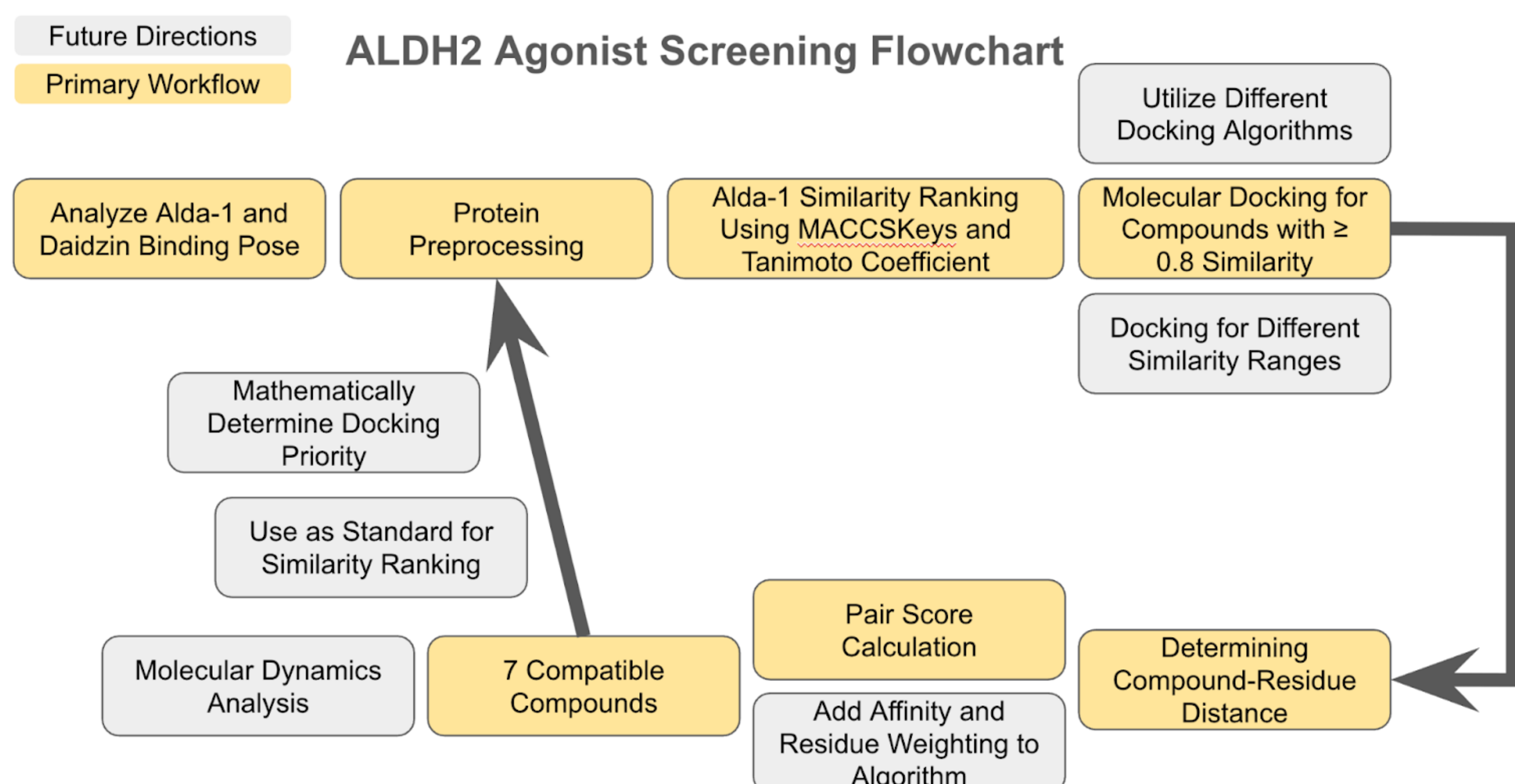


Figure 2: ALDH2 Agonist Screening Flowchart

1. Interaction of an Existing ALDH2 Agonist

Through entering the catalytic tunnel of ALDH2, Alda-1 interacts with 9 “whitelist” residues (Val120, Met124, Phe170, Leu173, Phe292, Phe296, Asp457, Val458, Phe459) to promote enzyme function. (Figure 3)

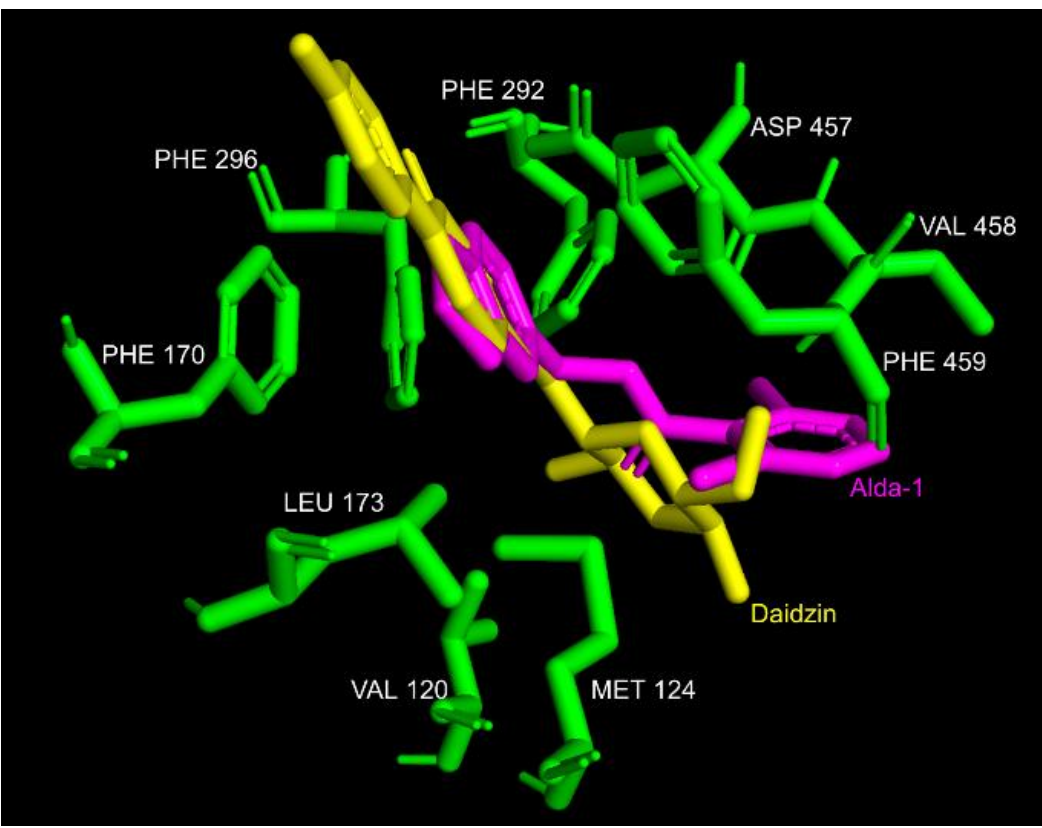


Figure 3: Interaction of Alda-1 and 9 ALDH2 residues

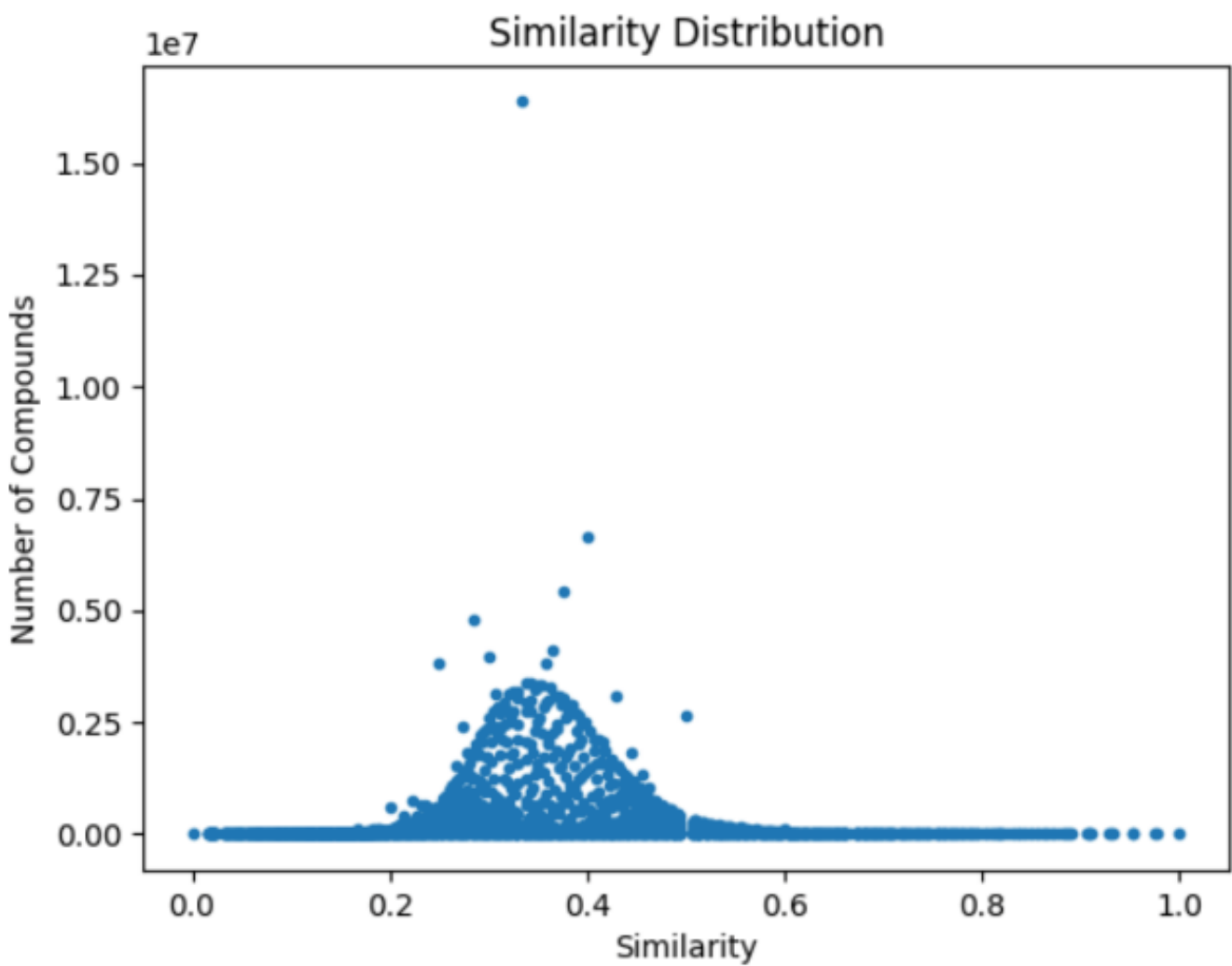


Figure 4: Distribution of Molecular Similarity between ZINC Compounds and Alda-1

2. Distribution of Molecular Similarity

After utilizing the Tanimoto Coefficient to calculate the similarity between 500M ZINC Database compounds and Alda-1, we plot the distribution with similarity as the x-axis and the number of compounds as the y-axis. As Figure 4 shows, a large majority of compounds range from 0.2 to 0.5. Note that only 9100 compounds show a similarity greater than 0.8.

3. Docking Position Analysis for >0.8 Similarity Compounds

After performing molecular docking using Autodock Vina, we plot the coordinates of the compounds and ALDH2 in a 3D space. We then clustered compound coordinates via DBSCAN ($\epsilon=1$) into 3 distinct groups after denoising, as shown in Figure 5. The cluster in green corresponds to the Alda-1 binding site on ALDH2; the cluster in blue is identical to the NAD⁺ binding site; the red cluster corresponds to the oligomerization domain of the enzyme, which would not be viable for substrate binding due to its role in forming tetramers. Detailed binding areas are depicted in Figure 6.

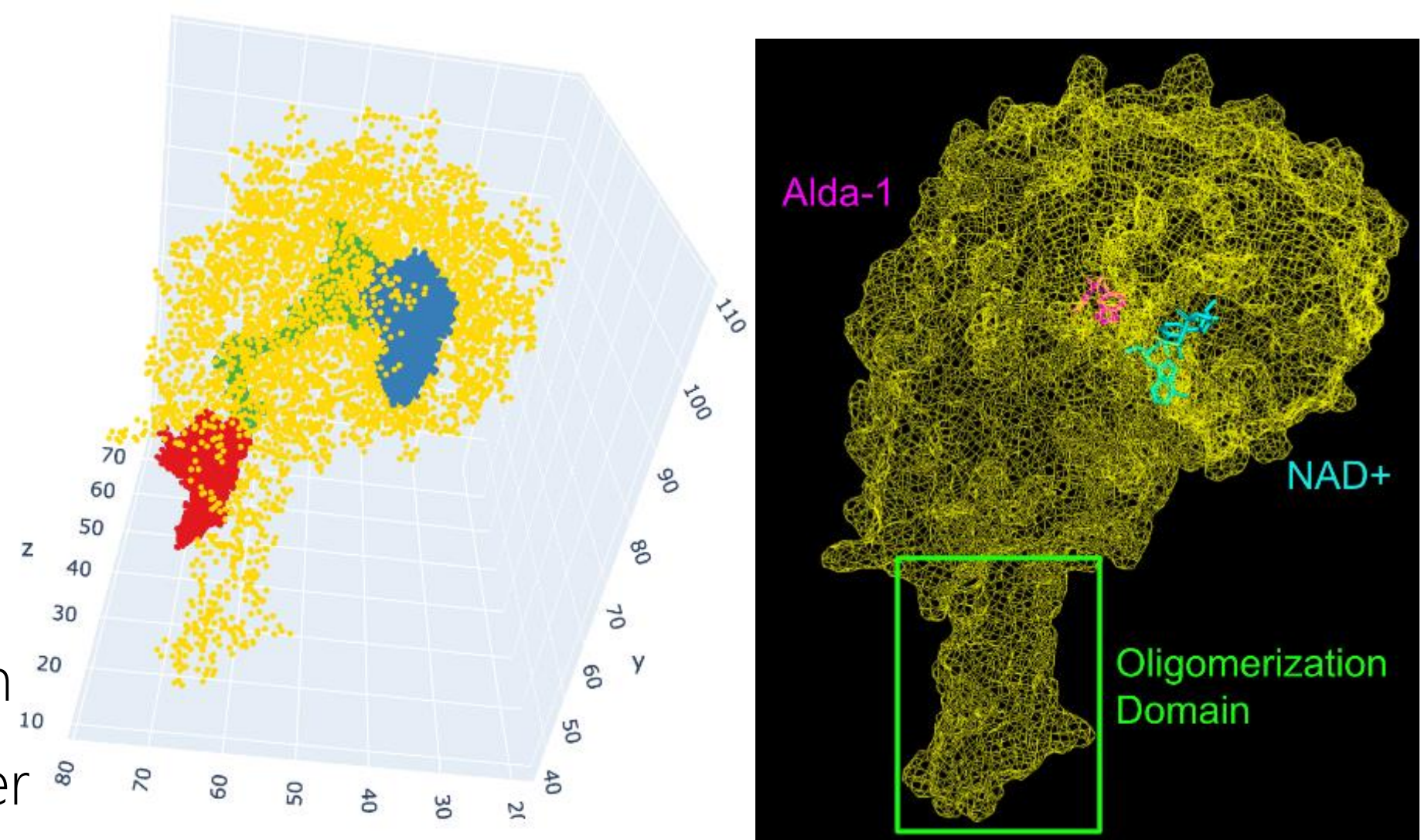


Figure 5: Docking Position of >0.8 Similarity Compounds (Red, blue, and green; yellow for ALDH2)

Figure 6: Binding Location of Alda-1 (Magenta), NAD⁺ (Turquoise), and the Oligomerization Domain (Green)

4. Post-docking Analysis and Individual Residue Interactions

We plotted the distribution of the binding affinities between >0.8 similarity compounds and ALDH2. (Figure 7) We then defined Pair Score as the number of whitelist residues within 4Å of the compounds, with only 7 compounds with a perfect score of 9. (Figure 8)

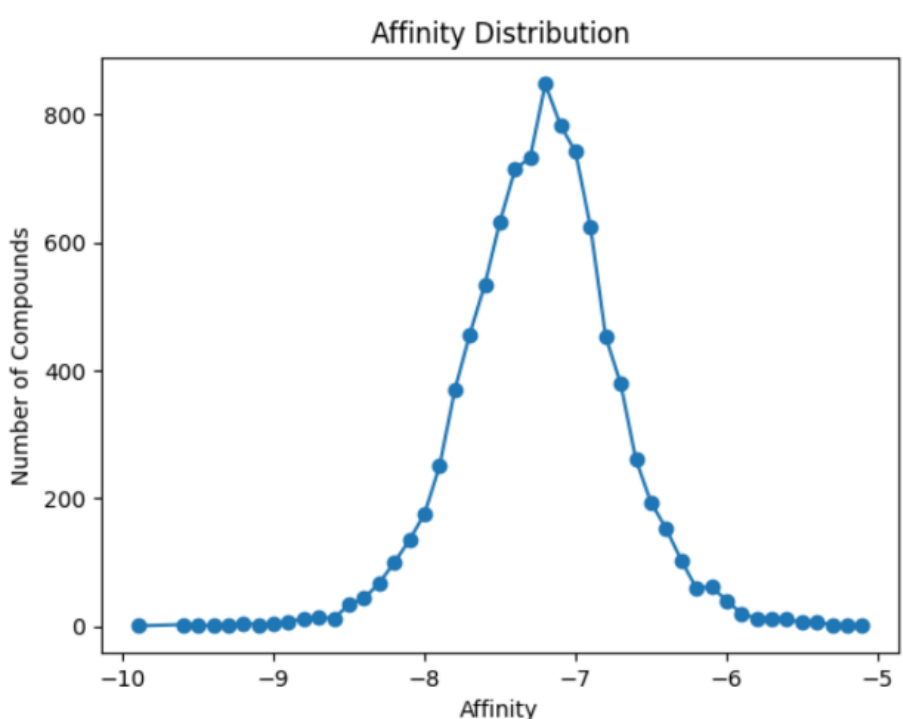


Figure 7: Affinity Distribution of >0.8 Similarity Compounds (Affinity to Number of Compounds)

Pair Score	Number of Compounds	Pair Score	Number of Compounds
0	8740	5	22
1	16	6	16
2	129	7	7
3	96	8	7
4	60	9	7

Figure 8: Pair Score Distribution

5. Post-Docking & Individual Residue Analysis

Below are the 7 ZINC Database compounds showing an identical docking position with Alda-1, suggesting potential in acting as ALDH2 agonists. Each compound has a perfect Pair Score of 9, interacting with all 9 whitelist residues (shown in white) and not interfering with the catalytic residues (shown in red).

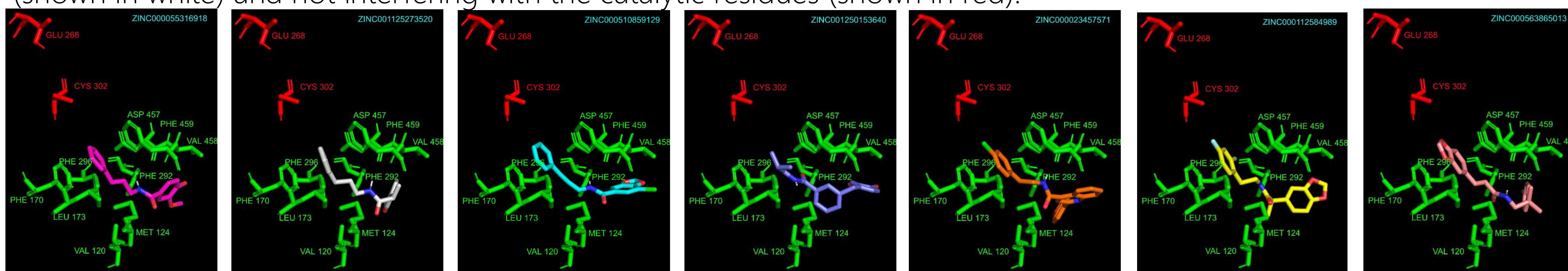


Figure9 ~ Figure 15: Docking Position of Pair Score = 9 compounds

Discussion

1. Viability of Similarity Ranking as an Agonist Screening Method

As shown in **Figure 4**, compounds with a >0.8 molecular similarity to Alda-1 only account for a small proportion of the total database. Therefore, our approach could significantly boost searching efficiency, but may compromise slightly on the structural diversity of compounds.

2. Improvement of Post-Docking Analysis and Pair Score

In our post-docking analysis, we only take the docking position and binding affinity of molecules into account in order to differentiate between agonists and inhibitors. However, our Pair Score does not consider binding affinity. We speculate that the range of affinity increases as similarity decreases, so as we expand our analysis to lower similarity molecules, we should add affinity as a parameter in our Pair Score function.

Moreover, interactions within the whitelist residues have been ignored in our approach. We assumed the probability of a compound as a potential agonist is proportional to the number of whitelist residues it interacts with. But in reality, different residues may have different levels of influence on enhancing the catalytic ability. We plan to add weights and biases to our Pair Score function to combat this issue.

3. Consideration of Additional Molecular Docking Algorithms

As of now, we have only utilized Autodock Vina as the sole molecular docking algorithm. However, we plan to compare different algorithms (iGEMDOCK, AlphaFold3) and their outputs to ensure accurate agonist prediction.

Future Prospects

The ultimate goal of this study is to address alcohol intolerance. Beyond computational screening, experimental validation is needed. After successfully isolating active ALDH2 enzymes, we will purify them, obtain the possible agonists, and test their activation effects based on enzyme kinetics. If these compounds yield successful agonists, they could guide drug development to improve alcohol metabolism.

References

1. Trott, O., & Olson, A. J. (2010). AutoDock Vina: improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *Journal of computational chemistry*, 31(2), 455-461.
2. Eberhardt, J., Santos-Martins, D., Tillack, A. F., Forli, S. (2021). AutoDock Vina 1.2.0: New Docking Methods, Expanded Force Field, and Python Bindings. *Journal of Chemical Information and Modeling*.